	Low	Moderate	High	Extreme
HA	+85.10	+62.84	+20.25	+11.98
SVR	+51.02	+31.24	+11.93	+7.98
VAR	+59.24	+39.83	+17.05	+8.11
DCRNN	+60.81	+17.46	+4.89	+4.98
STGCN	-1.06	+0.95	+1.22	+2.74
GWNet	-0.99	+2.44	+2.16	+3.72
STNorm	-4.53	-1.24	-1.21	+0.36
DSTAGNN	+81.88	+58.45	+20.53	+5.25
PDFormer	+83.11	+59.61	+19.42	+4.20
STWave	-4.86	-2.01	-1.50	+0.36
PatchTST	+70.02	+12.48	+7.45	+9.80
TimeMixer	+71.07	+17.32	+10.50	+9.85
STDN	+85.49	+70.43	+32.57	+5.55
MetaDG	+84.72	+66.55	+33.15	+12.43
HyperD	-0.70	+0.66	+0.24	+3.29
RAST	-0.65	+1.78	+2.45	+4.57
RBSRN MAE	**0.161**	**0.344**	**0.705**	**1.340**
Valid n	59,917	129,101	182,079	128,448

(a) Chengdu · inflow

	Low	Moderate	High	Extreme
HA	+86.85	+58.46	+15.35	+9.02
SVR	+53.48	+25.66	+8.83	+5.36
VAR	+62.29	+33.20	+12.24	+5.98
DCRNN	+59.90	+12.07	+2.32	+2.10
STGCN	-1.24	-0.31	+0.25	+1.71
GWNet	-2.31	+0.37	+0.53	+1.88
STNorm	-2.75	-1.77	-1.48	-0.18
DSTAGNN	+84.77	+55.60	+18.54	+4.60
PDFormer	+85.12	+54.90	+15.63	+3.51
STWave	-3.08	-2.17	-1.23	-0.60
PatchTST	+72.28	+9.07	+5.09	+7.79
TimeMixer	+73.16	+14.00	+8.28	+8.39
STDN	+87.35	+66.29	+25.88	+4.26
MetaDG	+86.66	+63.20	+28.82	+10.24
HyperD	-1.12	-0.59	-0.33	+1.11
RAST	-0.87	+1.05	+1.38	+2.71
RBSRN MAE	**0.139**	**0.380**	**0.765**	**1.246**
Valid n	58,257	150,159	182,619	135,869

(b) Chengdu · outflow

	Low	Moderate	High	Extreme
HA	+82.71	+39.22	+12.22	+13.20
SVR	+50.15	+17.74	+8.33	+7.62
VAR	+60.53	+25.30	+10.77	+7.76
DCRNN	+10.12	+2.19	+1.82	+3.50
STGCN	+2.23	+1.14	+1.29	+3.97
GWNet	+1.23	+1.10	+1.48	+3.51
STNorm	-0.02	-1.13	-1.29	+0.38
DSTAGNN	+78.69	+32.94	+8.82	+5.08
PDFormer	+80.57	+35.75	+10.33	+4.41
STWave	-0.88	-1.04	-1.22	+0.16
PatchTST	+57.96	+6.38	+5.28	+10.14
TimeMixer	+58.91	+10.59	+8.09	+10.18
STDN	+84.57	+53.93	+25.51	+15.28
MetaDG	+83.15	+49.75	+24.75	+15.09
HyperD	+5.08	+0.75	+0.17	+2.76
RAST	+1.06	+0.86	+1.36	+3.07
RBSRN MAE	**0.177**	**0.537**	**0.852**	**1.297**
Valid n	92,199	223,697	222,486	115,210

(c) Shanghai · inflow

	Low	Moderate	High	Extreme
HA	+84.41	+37.83	+12.12	+11.92
SVR	+53.45	+16.38	+8.16	+7.02
VAR	+64.81	+23.55	+10.79	+7.21
DCRNN	+10.84	+1.98	+1.67	+3.27
STGCN	+1.80	+0.64	+1.14	+2.83
GWNet	+0.51	+0.52	+1.18	+3.40
STNorm	-0.19	-1.25	-1.17	+0.21
DSTAGNN	+81.18	+32.03	+9.35	+3.45
PDFormer	+82.37	+34.29	+10.65	+3.79
STWave	-0.61	-1.23	-0.99	+0.07
PatchTST	+61.43	+5.92	+5.17	+9.37
TimeMixer	+62.13	+9.75	+8.10	+9.59
STDN	+85.65	+42.98	+14.01	+5.96
MetaDG	+84.71	+49.56	+28.05	+16.49
HyperD	+1.68	+0.34	+0.08	+0.98
RAST	+2.05	+1.07	+1.59	+3.90
RBSRN MAE	**0.158**	**0.553**	**0.870**	**1.482**
Valid n	84,381	216,920	227,734	110,618

(d) Shanghai · outflow

